Book 1 — Lecture 1 — Interlude 1 — Exercise 1

Plotting Four Introductory Functions

Prompt. Using a graphing calculator or a program like Mathematica, plot each of the following functions:

$$
\begin{aligned}
f(t) &= t^4 + 3t^3 - 12t^2 + t - 6, \\
g(x) &= \sin(x) - \cos(x), \\
\theta(\alpha) &= e^\alpha + \alpha \ln(\alpha), \\
x(t) &= \sin^2(t) - \cos(t).
\end{aligned}
$$

Notes Before Plotting

- $\theta(\alpha)$ is only defined for $\alpha > 0$, because $\ln(\alpha)$[1] needs a positive argument.
- $f(t)$ is a quartic polynomial, so it grows to $+\infty$ for large positive and negative t.
- $g(x)$ and $x(t)$ are bounded oscillatory functions built from sine/cosine terms.[2]

Example Plotting Windows

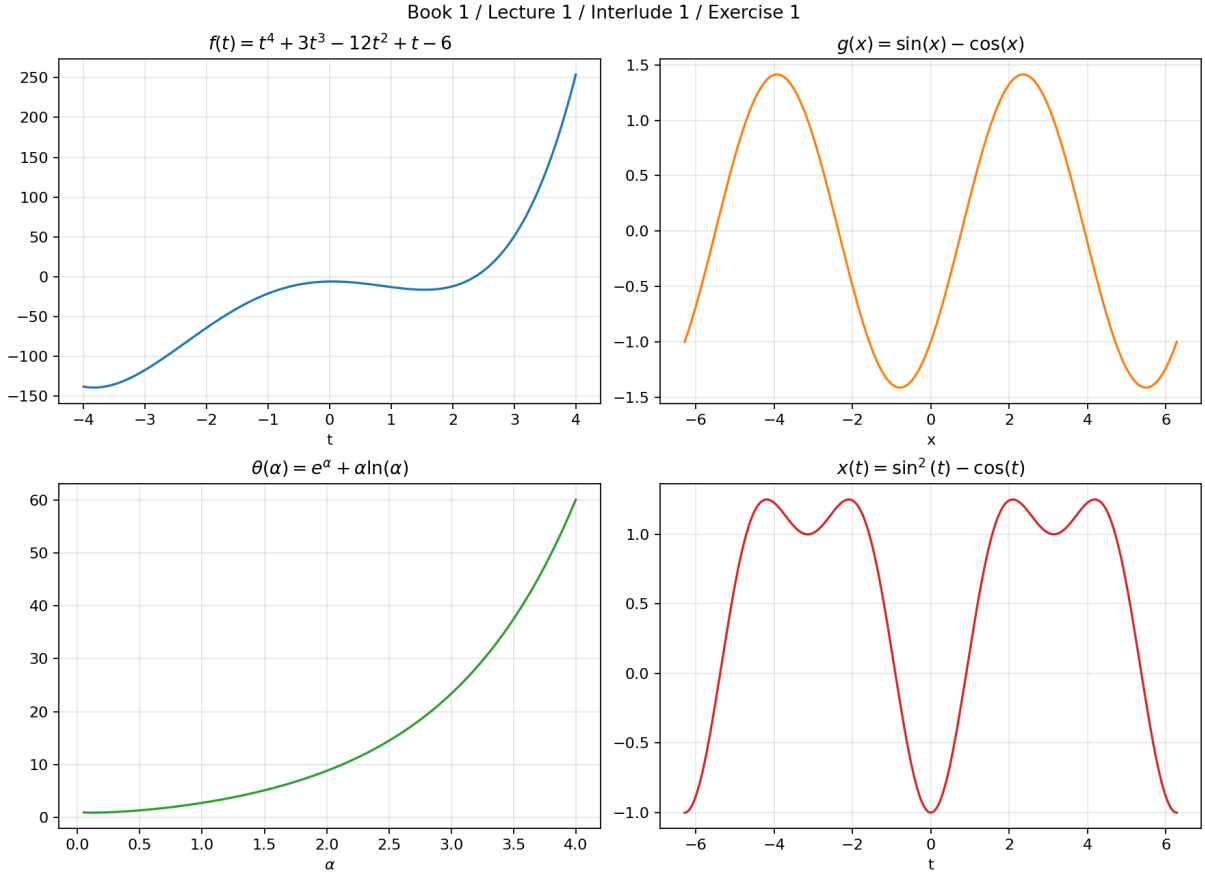
- $t \in [-4, 4]$ for $f(t)$
- $x \in [-2\pi, 2\pi]$ for $g(x)$
- $\alpha \in [0.05, 4]$ for $\theta(\alpha)$ (avoid $\alpha = 0$)
- $t \in [-2\pi, 2\pi]$ for $x(t)$

These are starting points; zooming in/out helps identify turning points and global behavior.

Result. A useful first pass is to use windows $t \in [-4, 4]$, $x \in [-2\pi, 2\pi]$, and $\alpha \in [0.05, 4]$.

Example. For $\theta(\alpha) = e^\alpha + \alpha \ln(\alpha)$, the logarithm term enforces $\alpha > 0$, so avoid $\alpha = 0$ in plotted ranges.

[1] ln is the natural logarithm (base e).

[2] sin and cos are trigonometric functions. Their outputs stay between -1 and 1.

Static Plot Figure